BIOT Module 7

Dhairya Bhatia

1. Introduction

By Module 6 I had a lot of evidence about my Alzheimer's genes, but it sat in separate tables: RNA and protein changes in one file, druggability in another, structures and risk scores in others. Each table answers its own question, but none of them can answer a question that crosses them, such as which drugs act on my genes, or which of my genes are shared with other brain diseases.

A knowledge graph puts all of this in one place. Genes, diseases, symptoms, pathways, drugs, variants and patients become nodes, and the links between them become edges, so a question becomes a walk through the graph. In this module I loaded my own evidence, plus public data around it, into a Neo4j database and queried it with Cypher.

This brings us to the main question of this module: once my Alzheimer's evidence is connected in one graph, which of my genes stand out, and what does the graph add that my separate tables could not show?

2. Disease and data choice, with biological interpretation

I stayed with Alzheimer's disease, so the graph is built around the same 15 genes I have followed since Module 2. Each gene node carries my own results: its RNA and protein change and GWAS strength (Module 2), its small-molecule and antibody scores (Module 3), and, for ITGAX and APLNR, the structure call (Module 4).

Layer	What it adds	Source
My genes and evidence	15 genes with Modules 2–4 results	my own outputs
Pathways	50 Reactome pathways	Open Targets [1], Reactome [2]
Drugs	5 drugs and clinical candidates on 3 of my genes	Open Targets [1]
Diseases	Alzheimer's and the two diseases sharing most of my genes	Open Targets [1]
Symptoms	140 HPO phenotypes of the three diseases	HPO [3] via Open Targets
Patients	200 people: real APOE genotypes, my Module 6 risk scores	Module 6 cohort [4]

I fetched the public layers with one script (`scripts/build_kg_data.py`) and saved them in `kg_data/`, so the graph can be rebuilt without the internet. The two related diseases were chosen by the data, not by me: Parkinson disease shares four of my genes (APOE, PECAM1, APOC1, HLA-DRB1) and schizophrenia shares three (APOE, NRXN3, APOC1), each with an Open Targets score of at least 0.1. The patients' case or control labels are the course's simulation from Module 6, so that layer shows the method, not real risk.

What the graph shows biologically:

- *My genes fall into the same groups as in Module 2, now confirmed by shared pathways:* fat transport (APOE, APOB and APOC1 meet in VLDL and chylomicron pathways), immune signalling (HLA-DRA and HLA-DRB1 in MHC class II antigen presentation) and blood-vessel and immune-cell surfaces (PECAM1 and ITGAX in integrin and vascular-wall interactions).
- *APOE links Alzheimer's to both related diseases,* which fits its known role across brain disorders.

3. Method

1. *Assemble:* build nodes and edges from my Modules 2–4 tables, the Open Targets data and 200 Module 6 patients (40 per ancestry group, sampled at random). Each gene was resolved to one Ensembl id through Open Targets, so the public data attaches to the right node.
2. *Load:* write the graph to my AuraDB Free instance with `UNWIND` batches; every node has its own label and a shared `:Entity` label with a unique name.

3. *Query in Cypher:* drugs acting on my genes (2 hops), a repurposing walk through related diseases that share a symptom with Alzheimer's (5 hops), and pathways shared by my genes.
4. *Prioritize:* PageRank in NetworkX on the graph without patients, fused with equal weight with three of my scores: the Module 2 score, the better of the two Module 3 druggability scores, and the Open Targets Alzheimer's score.
5. *Group and stratify:* modularity-based communities, and case rate and mean risk-score percentile by APOE genotype, in Cypher.

4. Results

The graph has 415 nodes and 329 relationships: 200 patients, 140 phenotypes, 50 pathways, 15 genes, 5 drugs, 3 diseases and 2 APOE variants.

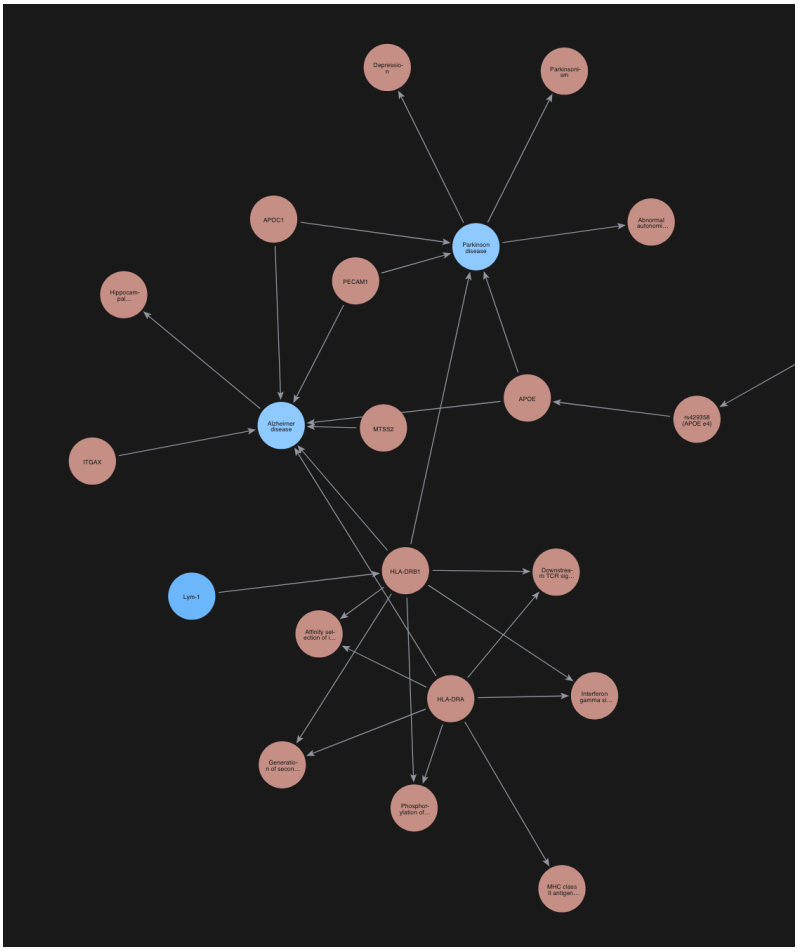


Figure 1: Part of my graph in Neo4j Explore. Alzheimer's and Parkinson disease (light blue) share APOE, APOC1, PECAM1 and HLA-DRB1; APOE carries the e4 variant rs429358; HLA-DRB1 and HLA-DRA share immune pathways such as MHC class II antigen presentation; Lym-1 (blue) is a drug candidate on HLA-DRB1. Arrows point from gene to disease, disease to symptom, and gene to pathway.

Drugs on my genes (2 hops): five drugs act on three of my genes. Mipomersen, an approved drug for high cholesterol, targets APOB. Apelin, a phase 2 candidate for heart failure and pulmonary hypertension, acts on APLNR. Three candidates act on HLA-DRB1: plovamer acetate (phase 2, multiple sclerosis), apolizumab (phase 2, blood cancers) and Lym-1 (phase 1, lymphoma). None is indicated for Alzheimer's.

Repurposing walk (5 hops): the three HLA-DRB1 candidates are reached through Parkinson disease, which shares depression and parkinsonism with Alzheimer's.

Table 1: Biomarker prioritization, top five of 15 and APLNR. Fused is the mean of the four percentile ranks.

Rank	Gene	PageRank	Module 2 score	Druggability	Open Targets AD	Fused
1	PECAM1	0.015	0.939	0.88	0.37	0.83
2	APOC1	0.010	0.949	0.25	0.57	0.73
3	ITGAX	0.011	0.912	0.59	0.23	0.68
4	HLA-DRB1	0.021	0.955	0.27	0.08	0.65
5	APOE	0.028	0.902	0.02	0.79	0.61
11	APLNR	0.009	0.902	0.68	0.02	0.47

Communities: six groups formed. Three match the biology above: a lipid group (APOE, APOB, APOC1 with the two APOE variants and mipomersen), an HLA group with its drugs, and a PECAM1–ITGAX group with their surface pathways. The other eight genes, which have few pathways, sit with the Alzheimer's node itself, and Parkinson disease and schizophrenia each form a group with their own symptoms.

Table 2: Patient stratification in Cypher (simulated case labels).

APOE genotype	People	Case rate	Mean risk-score percentile
e4/e4	5	40%	99
e3/e4	50	42%	86
e2/e4	3	100%	77
e3/e3	122	23%	41
e2/e3	20	40%	14

e4 carriers had a higher case rate than non-carriers in every ancestry group, from 37.5% against 25.0% in Europeans to 45.5% against 17.2% in South Asians.

5. Interpretation

Which genes stand out: four of my five Portfolio 1 shortlist genes are in the fused top five (PECAM1, ITGAX, HLA-DRB1, APOE). The graph adds APOC1 at second, because it has strong genetics, a high Open Targets score and links to both related diseases. PECAM1 tops the list because it scores well on every axis at once: strong omics, the best antibody score, shared vascular and immune pathways, and a link to Parkinson disease.

What the graph adds that the tables could not: three things. First, real drugs: mipomersen on APOB and apelin on APLNR are approved or in trials for other conditions, which my druggability scores could only predict. Second, links to other diseases: HLA-DRB1 reaches Parkinson's through shared symptoms, which no single table holds. Third, the groups my genes form, confirmed by shared pathways rather than by my reading of them.

Patient layer: the risk-score percentile rises with the number of e4 copies, from 41 in e3/e3 to 99 in e4/e4, as the score's design predicts. The case rates follow the same direction but are noisy, because some genotype groups are tiny (three e2/e4 people) and the labels are simulated.

6. Discordance and limitations

A gene where the graph and my earlier work disagree, APLNR: it led my Portfolio 1 shortlist as the best small-molecule target, but here it ranks 11th of 15. Its Open Targets Alzheimer's score is near zero (0.02) and it has only two pathways, so it sits at the edge of the graph. This shows a bias in knowledge graphs: well-studied genes have more links, so centrality rewards what is already known. APLNR's case rests on new evidence from my structure work, which the public graph does not yet hold.

Limitations:

1. *Annotation bias:* PageRank favours genes with many recorded pathways and drugs, such as APOB (highest PageRank), not necessarily the most important ones.
2. *Few drugs:* only five drugs reach my genes, and none for Alzheimer's, so the repurposing walk is narrow.
3. *Uneven symptoms:* schizophrenia has 103 HPO phenotypes against 11 for Alzheimer's, so one community is mostly its symptoms.

4. *Simulated patients:* case labels come from the course's Module 6 simulation, and 200 people is a small sample.
5. *A snapshot:* Open Targets changes over time; `kg_data/` records the version I used.

If I did this again, I would weight edges by evidence strength in PageRank, add ChEMBL bioactivity so that pre-clinical compounds count, and use real cohort labels if available.

References

[1] Ochoa, D., Hercules, A., Carmona, M., Suveges, D., Baker, J., Malangone, C., Lopez, I., Miranda, A., Cruz-Castillo, C., Fumis, L., Bernal-Llinares, M., Tsukanov, K., Cornu, H., Tsirigos, K., Razuvayevskaya, O., Buniello, A., Schwartzentruber, J., Karim, M., Ariano, B., . . . McDonagh, E. M. (2023). The next-generation Open Targets Platform: Reimagined, redesigned, rebuilt. *Nucleic Acids Research, 51*(D1), D1353–D1359. https://doi.org/10.1093/nar/gkac1046

[2] Milacic, M., Beavers, D., Conley, P., Gong, C., Gillespie, M., Griss, J., Haw, R., Jassal, B., Matthews, L., May, B., Petryszak, R., Ragueneau, E., Rothfels, K., Sevilla, C., Shamovsky, V., Stephan, R., Tiwari, K., Varusai, T., Weiser, J., . . . D'Eustachio, P. (2024). The Reactome Pathway Knowledgebase 2024. *Nucleic Acids Research, 52*(D1), D672–D678. https://doi.org/10.1093/nar/gkad1025

[3] Gargano, M. A., Matentzoglu, N., Coleman, B., Addo-Lartey, E. B., Anagnostopoulos, A. V., Anderton, J., Avillach, P., Bagley, A. M., Bakštein, E., Balhoff, J. P., Baynam, G., Bello, S. M., Berk, M., Bertram, H., Bishop, S., Blau, H., Bodenstein, D. F., Botas, P., Boztug, K., . . . Robinson, P. N. (2024). The Human Phenotype Ontology in 2024: Phenotypes around the world. *Nucleic Acids Research, 52*(D1), D1333–D1346. https://doi.org/10.1093/nar/gkad1005

[4] The 1000 Genomes Project Consortium. (2015). A global reference for human genetic variation. *Nature, 526*(7571), 68–74. https://doi.org/10.1038/nature15393